



Progressive Education Society's
MODERN COLLEGE OF ENGINEERING, Pune -05.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
MCA Department

PRACTICAL SUBMISSION RECORD- A.Y. 2024-25

Class: SYMCA	Div: B	Course Code: MCA01604	Batch: S2
Semester: III		Course Name: Data Science Laboratory	
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CO No: CO605.1		Assignment No: 02	

Title: Load the Titanic dataset, perform basic preprocessing (handle missing values, encode categorical variables), and train a Decision Tree classifier to predict the model. Plot the confusion matrix.

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

# Load Titanic dataset
titanic_url = 'https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv'
df = pd.read_csv(titanic_url)

# Basic preprocessing
# Drop columns that are not useful for prediction
cols_to_drop = ['PassengerId', 'Name', 'Ticket', 'Cabin']
df = df.drop(columns=cols_to_drop)

# Handle missing values
# Fill Age with median, Embarked with mode
if df['Age'].isnull().any():
    df['Age'] = df['Age'].fillna(df['Age'].median())
if df['Embarked'].isnull().any():
    df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])

# Encode categorical variables
# Sex and Embarked
for col in ['Sex', 'Embarked']:
    df[col] = pd.Categorical(df[col]).codes

# Define features and target
X = df.drop('Survived', axis=1)
y = df['Survived']
```



```
# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)

# Train Decision Tree classifier
clf = DecisionTreeClassifier(random_state=42)
clf.fit(X_train, y_train)

# Predict
y_pred = clf.predict(X_test)

# Plot confusion matrix
cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=clf.classes_)
disp.plot(cmap=plt.cm.Blues)
plt.title("Titanic Decision Tree Confusion Matrix")
plt.show()
```

Output:

